Trigonometric Function
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

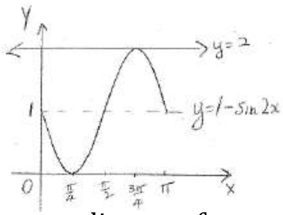
1 (i) Express $5\,cot^2\,x - 2cosecx + 2$ in terms of $cosecx$. 1

$5(Cosec^2x - 1) - 2Cosecx + 2$
$= 5Cosec^2x - 5 - 2Cosecx + 2$
$= 5Cosec^2x - 2Cosecx - 3$

(ii) Hence, solve the equation $5\,cot^2\,x - 2cosecx + 2 = 0$ for $0 \leq x \leq 2\pi$. 2

$5Cosec^2x - 2Cosecx - 3 = 0$
$5Cosec^2x - 5Cosecx + 3Cosecx - 3 = 0$
$5Cosecx(Cosecx - 1) + 3(Cosecx - 1) = 0$
$(Cosecx - 1)(5Cosecx + 3) = 0$
$Cosecx = 1$ $\quad$ $Cosecx = -\frac{3}{5}$
$x = \frac{\pi}{2}$ $\quad$ No solution
$\therefore x = \frac{\pi}{2}$ only solution

2 (i) Sketch the curve $y = 1 - sin2x$ for $0 \leq x \leq \pi$. 1

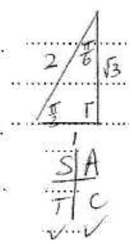


(ii) On the sketch above draw $y = 2$ and write down the co-ordinates of any point(s) of intersection between $y = 1 - sin2x$ and $y = 2$. 1

point of intersection $\left(\frac{3\pi}{4}, 2\right)$

3 Solve $2\,sin^2\,x - 3sinx - 2 = 0$ for $0 \leq x \leq 2\pi$. 2

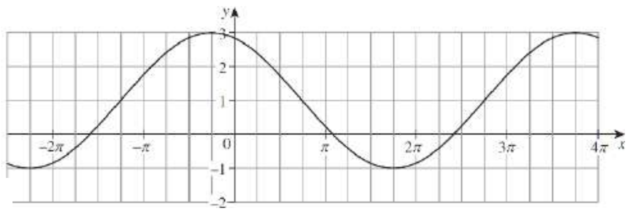
$(2sinx + 1)(sinx - 2) = 0$
$\therefore sinx = 2$ $\quad$ $sinx = -\frac{1}{2}$
no solution $\quad$ $\therefore x = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$
$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6}$



4 What is the maximum value of $y = 2\,sin\left(\frac{x}{3}\right) + 1$ and the value of x occurring in the domain $[0, 2\pi]$? 2

The amplitude is 2; hence, the maximum value of $y = 2\sin\left(\frac{x}{3}\right)+1$ is 3.
When $y = 3$: $\quad 3 = 2\sin\left(\frac{x}{3}\right)+1 \quad 1 = \sin\left(\frac{x}{3}\right)$
$\frac{x}{3} = \frac{\pi}{2} \quad x = \frac{3\pi}{2}$

5 The graph is a function of the form $y = kcos\big(a(x + b)\big) + c$. Find the equation of the function. 3



Vertical dilation:
amplitude $= \frac{3-(-1)}{2} = 2$ $\quad$ Therefore, $k = 2$.
Vertical translation:
The centre line is $y = 1$. This means the graph has undergone a vertical shift of 1. Therefore, $c = 1$.
Horizontal dilation:
Notice the period is 4π. Hence: $\frac{2\pi}{a} = 4\pi \Rightarrow a = \frac{1}{2}$ $\quad \therefore a = \frac{1}{2}$
Horizontal translation:
The horizontal translation is to the left by $\frac{\pi}{4}$.
Therefore, $b = \frac{\pi}{4}$.
The equation of the function is $y = 2\cos\left(\frac{1}{2}\left(x + \frac{\pi}{4}\right)\right) + 1$.

6 (i) Show that $(cosec^2 A - 1)\, sin^2 A = cos^2 A$. 2

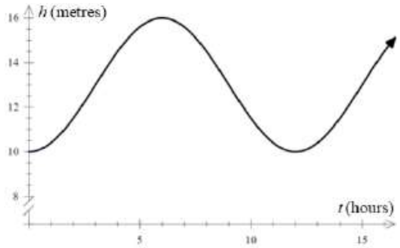
$$LHS = \left(\operatorname{cosec}^2 A - 1\right)\sin^2 A = \cot^2 A \sin^2 A$$
$$= \frac{\cos^2 A}{\sin^2 A} \times \sin^2 A = \cos^2 A = RHS$$

(ii) Hence, or otherwise, solve $(cosec^2 A - 1)\, sin^2 A = \frac{3}{4}$ for $-\pi \le A \le \pi$. 3

$$\cot^2 A \times \sin^2 A = \frac{3}{4}$$
$$\cos^2 A = \frac{3}{4} \qquad \cos A = \pm\frac{\sqrt{3}}{2}$$
$$\text{Related angle } = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$
$$A = \frac{\pi}{6}, \frac{5\pi}{6}, -\frac{\pi}{6}, -\frac{5\pi}{6}$$

7 The rise and fall in sea level in The Suez Canal, due to tides, can be modelled by the cosine function below: $h(t) = Acos(bt) + d$

At 8am it is low tide, and the channel is 10m deep. At 2pm it is high tide, and the channel is 16m deep. A ship needs 11.5m of water to sail.



(i) Find the values of A, b and d. 2

$P = \frac{2\pi}{b}$ Period 12h (8am to 8pm)

$12 = \frac{2\pi}{b}$

$b = \frac{\pi}{6}$

max (high tide) = 16m

min (low tide) = 10m

$\frac{10+16}{2} = \frac{26}{2} = 13$

$\therefore$ centre $= 13$

$\therefore d = 13$

$\therefore$ amplitude $= 3$

$A = 3$

$\therefore \ h(t) = -3\cos\left(\frac{\pi t}{6}\right) + 13$

$\therefore \ A = -3 \quad b = \frac{\pi}{6} \quad t = 13$

(ii) Between what times in the first day can the ship sail? 2

$11.5 = -3\cos\left(\frac{\pi t}{6}\right) + 13$

$-1.5 = -3\cos\left(\frac{\pi t}{6}\right)$

$0.5 = \cos\left(\frac{\pi t}{6}\right)$

$\frac{\pi t}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$

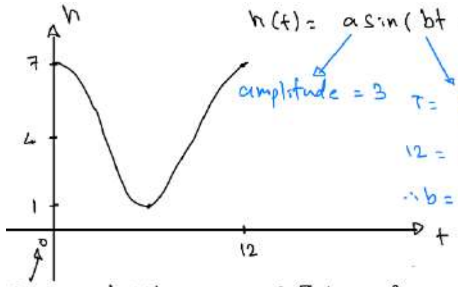
$t = 2, 10$

$\therefore$ between 10am & 6pm

8 Solve $sin\left(x + \frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$ for $0 \le x \le 2\pi$. 2

$$\sin\left(x + \frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$$
$$x + \frac{\pi}{6} = \frac{\pi}{3} \quad \text{acute related angle}$$
$$= \frac{4\pi}{3}, \frac{5\pi}{3}$$
$$x = \frac{7\pi}{6}, \frac{3\pi}{2}$$

9 In a tidal river, the time between high tides is 12 hours. The average depth of water at a point the river is 4 metres and at high tide, the depth is 7 metres. The depth of water $h(t)$ metres, at this point is given by $h(t) = asin(bt + c) + d$ where t is the number of hours after noon. At noon there is high tide. Find the values of positive real numbers a, b, c and d. 3



$h(t) = a\sin(bt + c) + d$

amplitude $= 3$

$T = \frac{2\pi}{b}$

$12 = \frac{2\pi}{b}$

$\therefore b = \frac{\pi}{6}$

$d =$ centre

$d = 4$

$h(t) = 3\sin\left(\frac{\pi}{6}t + c\right) + 4$

when $t = 0, \quad h = 7$

$7 = 3\sin\left(\frac{\pi}{6} \times 0 + c\right) + 4$

$1 = \sin c \quad \therefore c = \frac{\pi}{2}$

$\therefore \ h(t) = 3\sin\left(\frac{\pi}{6}t + \frac{\pi}{2}\right) + 4$

10 For what values of x, in the interval $0 \leq x \leq \frac{\pi}{4}$, does the line $y = 1$ intersect the graph of $y = 2sin4x$? 2

$2\sin(4x) = 1 \qquad \sin(4x) = \frac{1}{2}$

To solve for $0 \leq x \leq \frac{\pi}{4}$, we need to solve the above trigonometric equation for $0 \leq 4x \leq \pi$.

$\therefore 4x = \frac{\pi}{6}, \frac{5\pi}{6}. \qquad \therefore x = \frac{\pi}{24}, \frac{5\pi}{24}.$

11 Solve $|1 - 2\,cos^2 x| = 1$ for $0 \leq x \leq 2\pi$. 2

$1 - 2\cos^2 x = 1$

$2\cos^2 x = 0$

$\cos^2 x = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2}$

$1 - 2\cos^2 x = -1$

$2\cos^2 x = 2$

$\cos^2 x = 1$

$\cos x = \pm 1$

$x = 0, \pi, 2\pi$

$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$

Trigonometric Function
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 What is the solution to the equation $cos2x = \frac{1}{2}$ in the domain $[-\pi, \pi]$? 1

$\cos 2x = \frac{1}{2}$ $\quad -2\pi \le 2x \le 2\pi$

related $\angle = \frac{\pi}{3}$

$\therefore\ 2x = \frac{\pi}{3},\ 2\pi - \frac{\pi}{3},\ -\frac{\pi}{3},\ -2\pi + \frac{\pi}{3}$

$2x = \frac{\pi}{3},\ \frac{5\pi}{3},\ -\frac{\pi}{3},\ -\frac{5\pi}{3}$ S A ✓ T C ✓

$x = \frac{\pi}{6},\ \frac{5\pi}{6},\ -\frac{\pi}{6},\ -\frac{5\pi}{6}$

2 Compared with the graph of $y = cosx$ the graph of $y = cos\frac{x}{2}$ has: 1

A. half the amplitude but the same period

B. the same amplitude and half the same period

C. double the amplitude and the same period

D. the same amplitude and double the period

3 Solve the equation $tan^2 x = 3$ for $0 \le x \le 2\pi$. 2

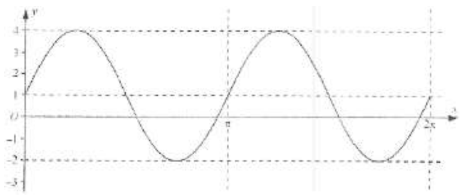
$(\tan x)^2 = 3$

$\tan x = \pm\sqrt{3}$ S ✓ A ✓ T ✓ C ✓

related angle $= \frac{\pi}{3}$

$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

4 The diagram shows the graph of $y = asin(bx) + c$ for $0 \le x \le 2\pi$, where a, b and c are positive integers. Find the values of a, b and c. 2



$a = 3$

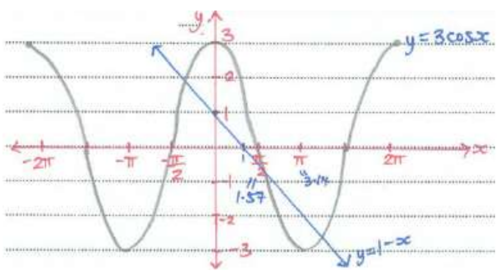
$c = 1$

$T = \pi$

$\therefore \frac{2\pi}{b} = \pi$

$\therefore b = 2$

5 (i) Sketch $y = 3cosx$ in the domain $-2\pi \le x \le 2\pi$. 2



(ii) Hence or otherwise, how many solutions are there to the equation $3cosx = 1 - x$? 1

Sketch $y = 1 - x$
y-intercept: $y = 1$
x-intercept: $x = 1$

3 solutions (as there are 3 points of intersection)

Amplitude is multiplied by 3 and the period is halved

6 The function $f(x) = sinx$ is transformed to $g(x) = 3sin2x$. Describe in words how both the amplitude and period change in this transformation. 2

7 Solve $3tan\theta - cot\theta = 5$ for $0^o \leq \theta \leq 360^o$. Give your answer to the nearest degree. 3

$3\tan\theta - \cot\theta = 5$

$\cot\theta = \frac{1}{\tan\theta}$

$3\tan^2\theta - 1 = 5\tan\theta$

$3\tan^2\theta - 5\tan\theta - 1 = 0$

$\tan\theta = \frac{5 \pm \sqrt{25+12}}{6} = \frac{5 \pm \sqrt{37}}{6}$

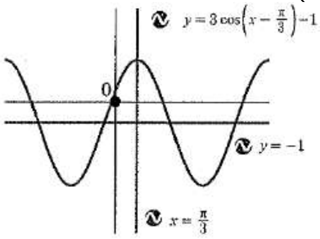
$\tan\theta = \frac{5+\sqrt{37}}{6}$ | $\tan\theta = \frac{5-\sqrt{37}}{6}$

$\theta = 62^o,\ 180^o + 60^o$ | $\theta = -10.22^o + 360^o = 350^o$

$\theta = 62^o, 242^o$ | $\theta = -10.22^o + 180^o = 170^o$

8 Identify the various transformations of the standard functions to $y = 3\,cos\left(x - \frac{\pi}{3}\right) - 1$ and hence graph it. 2

The graph of $y = \cos x$ dilated vertically with factor 3, shifted horizontally $\frac{\pi}{3}$ units to the right and shifted vertically 1 unit down.



9 Write down a sequence of transformations that will transform $y = cosx$ into:

(i) $y = 4cosx + 3$ The graph of $y = \cos x$ is stretched vertically with factor 4, then shifted up 3 units. 1

(ii) $y = \frac{5}{3}cos\left(2\left(x + \frac{\pi}{6}\right)\right)$ The graph of $y = \cos x$ is stretched vertically with factor $\frac{5}{3}$, then stretched horizontally with factor $\frac{1}{2}$, then shifted left $\frac{\pi}{6}$ units. 2

(iii) $y = -\frac{7}{3}cos\left(2x - \frac{3\pi}{2}\right)$ The graph of $y = \cos x$ is reflected in the x-axis, stretched vertically with factor $\frac{7}{3}$, shifted right $\frac{3\pi}{2}$ units and then stretched horizontally with factor $\frac{1}{2}$; or 2

the graph of $y = \cos x$ is reflected in the x-axis, stretched vertically with factor $\frac{7}{3}$, then stretched horizontally with factor $\frac{1}{2}$ and shifted right $\frac{3\pi}{4}$ units.

10 Solve $2\,cos\left[2\left(x - \frac{\pi}{6}\right)\right] = 1$ in the domain $[0, 2\pi]$. 3

$2\cos\left[2\left(x - \frac{\pi}{6}\right)\right] = 1$

$\cos\left[2\left(x - \frac{\pi}{6}\right)\right] = \frac{1}{2}$

Let $2\left(x - \frac{\pi}{6}\right) = \theta$

$\cos\theta = \frac{1}{2}$ and cos $+ve$ 1st & 4th quad and in repetitions of these positions

$\theta = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$

If $\theta = 2\left(x - \frac{\pi}{6}\right)$

$2\left(x - \frac{\pi}{6}\right) = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$

$x - \frac{\pi}{6} = -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}$

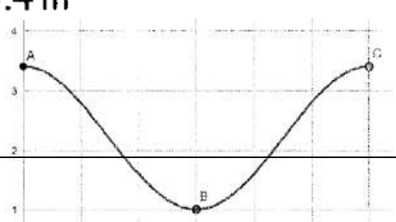
$x = 0, \frac{\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi, \cancel{\frac{7\pi}{3}}$

$x = 0, \frac{\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi$

11 The height, $h(t)$ metres, that the sea water reaches up one particular wooden stump at Bondi Pier during a 24-hour period can be modelled by the rule $h(t) = 2.2 + 1.2\,cos\left(\frac{\pi t}{12}\right)$ where t is the number of hours after 5 am and $0 \leq t \leq 24$.

$h(0) = 3.4$ m

(i) Find the height of the water on the stump at 5 am. 1



(ii) Sketch the graph of $y = h(t)$ for $0 \leq t \leq 24$. 2

(iii) What will be the time when the height of the water drops to 1.6 metres for the first time? 2

$$2.2 + 1.2\cos\left(\frac{\pi t}{12}\right) = 1.6 \Rightarrow \cos\left(\frac{\pi t}{12}\right) = -\frac{1}{2} \Rightarrow \left[\begin{array}{l} \frac{\pi t}{12} = \frac{2\pi}{3} \\ \frac{\pi t}{12} = \frac{4\pi}{3} \end{array}\right. \Rightarrow \left[\begin{array}{l} t = 8 \\ t = 16 \end{array}\right.$$

$$\therefore t = 8$$

(iv) For how many hours will the height of the water be above 1.6 metres during a 24-hour period? $24 - (16 - 8) = 16$ hr 1